Litigation as a 'win-win' game: A Theoretical Perspective and Empirical Evidence from Brazilian Labor Courts

Luciana L. Yeung
LucianaY@insper.edu.br
Insper

Antônio B. C. Morales
antoniobcm2@insper.edu.br
Insper

August 12, 2026

Abstract

Employers in Brazilian labor courts are ordered to pay something in roughly nine of every ten adjudicated claims, and a composition is available at any time at a fraction of what a judgment costs them. Yet large employers decline to compose, claim after claim, for years. We show that this is the equilibrium of a repeated game in which a long-lived defendant faces a continuing flow of short-lived claimants who differ in the personal cost of carrying a claim. The defendant is not buying a cheaper resolution of the present claim; it is buying a public record of refusing to compose, which lengthens the expected duration of any claim brought against it and thins the flow of claims it will face in future. The mechanism requires neither divergent beliefs about the merits nor private information. Three results carry the argument. A reputation for toughness deters entry if and only if the ratio of the two values at stake is smaller than the ratio of the two delays, each measured as a discounted annuity, a condition that is automatic when compositions price at the claimant's reservation value. Because claimants differ, deterrence is always *partial* and the filing rate is bounded away from zero at every reputation the defendant can attain, so the stationary outcome is persistent litigation rather than its disappearance. And permanent toughness requires the defendant's patience to exceed a threshold that falls with the responsiveness of entry and with the speed at which the market's record forms. We take the model to approximately 130,000 decisions of Brazil's largest regional labor court (TRT-2, São Paulo) for 2001–2017, obtained by text mining. The corpus establishes the institutional configuration the model assumes: court-fee waivers granted essentially automatically and held by 84.7% of claimants, no exposure of the losing claimant to the employer's attorney fees, and contingent representation, which together produce a lottery with no losing ticket and guarantee participation at any degree of risk aversion. It also establishes the constraint that forces our definition of reputation, since an employer that disputes everything still loses most of what it disputes and cannot build a record of victories. We do not report a calibration: composition rates are not observable in a corpus of decisions, and we set out what would be needed to measure the model's remaining objects.

Keywords: Litigation; Settlement; Labor Courts; Repeated Games; Reputation; Risk Aversion; Text Mining.

JEL Classification: K41, C72, C73, J52, K31.

1 Introduction

Why would a party that is ordered to pay in the large majority of the claims it defends decline, claim after claim and year after year, to compose those claims at a fraction of what a judgment costs it? That question defines the puzzle of this paper. The answer we give is that the employer is not buying a cheaper resolution of the present claim. It is buying a public record of refusing to compose, and that record lengthens the expected duration of every claim that might be brought against it in future.

Litigation rates in Brazil are high by any account. Ramseyer and Rasmusen (2013) measure suits filed per 100,000 people in several developed countries and report 5,806 for the United States, 3,681 for England and 1,768 for Japan; Wollschläger (1998) reports 12,300 for Germany and 11,120 for Sweden. The Brazilian National Council of Justice reports that at the end of 2017 there were 80.1 million cases pending before Brazilian courts and that roughly 29 million new cases were filed during that year (Conselho Nacional de Justiça, 2018). With a population near 208 million, that is about 0.38 pending cases per inhabitant and roughly 14,000 new cases per 100,000 inhabitants per year.

Two cautions belong with those comparisons and we state them rather than let the numbers do rhetorical work they cannot support. The figures usually quoted for other jurisdictions are *flows* of civil suits, whereas the Brazilian stock of pending cases is often quoted alongside them; measured on a common flow basis Brazil is of the same order as Germany rather than a multiple of it. And the Brazilian aggregate includes tax enforcement proceedings, which are administrative debt collection rather than civil litigation in the sense of those studies. The order of magnitude has not changed since: the 2025 report records 80.6 million cases pending (Conselho Nacional de Justiça, 2025). We stop the data analysis in 2017 because the labor law reform of that year altered several of the parameters our model identifies as central, and produced a sharp downward break in filings.

Labor courts are the sharp end of this. Pinheiro (2000), one of the first economic and empirical analyses of the Brazilian judiciary, surveyed approximately 600 firms of all sizes about their experience in court over the preceding decade. Table 1 reproduces the counts.

Table 1: Number of Lawsuits Faced in Courts ($n = 600$ companies)

Matter of Dispute	As plaintiff	As defendant	Total
Labor	1,904	105,753	107,657
Tributary	8,394	3,027	11,421
Commercial	11,071	3,096	14,167
Intellectual Property	199	23	222
Consumer's Rights	37	920	957
Environmental	12	68	80

Source: Pinheiro (2000).

In comparison with other fields of law the number of labor cases is staggering: the second most litigious area, commercial law, comprises roughly 13% of the labor total. Official statistics point the same way. In 2016 there were 1,721 new labor cases for every 100,000 people, a total of 4.3 million, and labor courts accounted for 5.4 million pending cases, third behind state and federal courts, both of which handle a far wider range of subject matter.

What we do. We build a repeated game in which one long-lived employer faces, each period, a fresh cohort of potential claimants who differ in a private participation cost: the time, stress, foregone opportunities and perceived risk of retaliation involved in keeping a claim alive. The employer's public record is the share of past claims it took to judgment rather than composing, and a claimant deciding whether to file reads that record as the probability that her own claim will be adjudicated rather than composed. Since adjudication is slow and conciliation is fast, a harder record raises the expected cost of waiting without raising the expected value of the claim, and the claimants with the highest participation cost drop out.

Three results carry the argument.

First, a reputation for toughness deters entry if and only if the ratio of the two values at stake is smaller than the ratio of the two delays, each measured as a discounted annuity (Lemma 5.2). The condition is easy to satisfy when adjudication is slow relative to conciliation, and it holds automatically when compositions are priced at the claimant's reservation value.

Second, because claimants differ, deterrence is always *partial*. However tough the employer becomes, the claimants with the lowest participation cost still sue, the filing rate is bounded away from zero at every attainable reputation, and the stationary outcome is persistent litigation rather than its disappearance (Lemma 5.3). This matters more than it may appear. A model in which deterrence is a threshold predicts an absorbing state in which filings stop forever, which is the opposite of the phenomenon to be explained; heterogeneity is what allows a stationary regime of heavy litigation to be an equilibrium rather than a transition.

Third, permanent toughness requires the employer's patience to exceed a threshold δ^* that falls with the responsiveness of entry and with the speed at which the market's record forms (Proposition 5.2), and it beats composing every claim when the deterrence dividend of Definition 5.8 exceeds $1/(1-\kappa)$, where κ is the static cost of toughness (Lemma 5.5 and Corollary 5.4). The two conditions are distinct and neither implies the other.

We treat the two extensions as competing accounts of the same margin rather than as components of one account, and Section 6.2 asks the corpus about each separately. The answers differ: claimant risk aversion, taken alone, predicts *more* composition rather than less, so it cannot be the explanation; the reputational mechanism reproduces the two features of the data that the alternatives do not, but its central prediction is not observable in a corpus of decisions. A model in which the two operate jointly is the natural next step and is not attempted here.

The mechanism requires neither divergent beliefs about the merits nor private information about them. In this respect it belongs with a small literature on settlement failure under complete information, discussed in Section 2, and it departs from the canonical reputation models in the object that does the deterring: entry here is not one binary decision by one entrant but a flow of short-lived players who differ in the cost of entering.

What we do not claim. Two clarifications bound what the empirical part of the paper supports, and we make them at the outset.

We do not test the selection hypothesis of Priest and Klein (1984). Our outcome variable counts a ruling as favourable to the claimant when it grants at least one of the cumulated claims, which in Brazilian labor practice is close to a mechanical event; and, as Shavell (1996) and Klerman and Lee (2014) have established, essentially any plaintiff win rate is consistent with rational selection. We use the observed rate to fix one parameter

and draw no inference from its distance to fifty percent.

We do not report a calibration. Cases resolved by conciliation produce no reasoned judgment and are therefore absent from a corpus of decisions by construction, so the composition rate, which is the observable counterpart of our model's state variable, cannot be measured from these data. Nor is the extraction yet validated. Section 6 states what the corpus does establish, which is the institutional configuration the model assumes and the constraint that forces our definition of reputation, and sets out in order what would be required to measure the rest.

The paper proceeds as follows. Section 2 positions the argument in the literature. Section 3 describes the data and the extraction. Section 4 develops the static base model. Section 5 adds claimant risk aversion and employer reputation and treats them as alternative accounts of the same margin. Section 6 confronts the model with the corpus. Section 7 concludes.

2 Related Literature

According to Miller (1994), "of all fields in which economic analysis has contributed to our understanding of legal rules and institutions, few have been as fruitful as that of civil procedure" (p. 303). The literature asks, in general terms, what generates litigation as opposed to cooperative settlement. We organise the strands that bear on our argument, and state where we sit with respect to each.

2.1 Why cases fail to settle

The oldest answer is *divergent expectations*. In Landes (1971), Gould (1973) and Posner (1973), settlement fails when the parties' subjective probabilities of success differ by more than the joint cost of trial. Priest and Klein (1984) build the selection hypothesis on that foundation and conclude that, where gains and losses from litigation are equal for the parties, individual maximisation creates a strong bias toward a plaintiff success rate of fifty percent regardless of the substantive standard of law.

Shavell (1982) offers the foundational treatment of the decision to sue, settle or proceed to trial, and shows that positive litigation costs create a range of settlement values both parties prefer to trial, a range that widens as costs rise. Cooter and Rubinfeld (1989) obtain related results under rational choice assumptions. The second answer is *asymmetric information*: P'ng (1983), Bebchuk (1984), Reinganum and Wilde (1986) and Spier (1992) derive settlement failure from private information about the merits or about the opponent's type, and Daughety and Reinganum (2005) and Spier (2007) survey the field.

Eisenberg (1990) notes two limitations of the selection hypothesis: it does not apply when the extent rather than the existence of liability is in dispute, and it applies only when the stakes are the same for both litigants, "a condition that is not satisfied where one party (such as an institutional defendant) worries about the effect that the outcome of the case will have on other litigation, whereas the other party (such as the ordinary tort plaintiff) does not" (p. 339). That is exactly the asymmetry of labor litigation, and it is the asymmetry our model imposes. Subsequently Shavell (1996) proved that any frequency of plaintiff victory is consistent with rational selection, and Klerman and Lee (2014) established the precise conditions under which the fifty percent result holds. We take that literature to have settled the matter, and accordingly build no part of our argument on the distance between an observed win rate and fifty percent.

Closest to us in spirit is a smaller literature in which settlement fails under *complete* information. Hubbard (2018) shows that even with complete information and fully rational parties, a defendant may find stalling optimal outside of litigation, leaving the plaintiff no alternative but to file. Our mechanism shares that feature and differs in where the delay bites: in Hubbard (2018) the defendant delays before the suit, forcing the plaintiff into court, whereas here the defendant's refusal to compose lengthens proceedings already underway and thereby deters the *next* claimant.

2.2 Repeat players and reputation

Galanter (1974) argues that parties who litigate frequently, such as large corporations, develop systematic advantages over "one-shotters" such as individual plaintiffs: they accumulate expertise, maintain ongoing relationships with lawyers and courts, and can adopt long-run strategies that would be irrational for a party facing a single dispute. Our model formalises the last of these. Its structure is that of the entry-deterrence literature, in which a long-lived player facing a sequence of short-lived opponents invests in a public record in order to alter their entry decisions (Fudenberg & Levine, 1989; Kreps & Wilson, 1982; Mailath & Samuelson, 2006; Milgrom & Roberts, 1982). Che and Yi (1993) study the role of precedent in repeated litigation and Daughety and Reinganum (1999) the defendant's incentive to suppress information across cases.

Our departure from that literature is the entry margin. In the canonical chain-store setting, entry is a binary decision by a single entrant and deterrence is complete. Litigation is not like that: a large employer faces a continuing flow of potential claimants who differ in what suing costs them personally. Deterrence is therefore always partial, the reputational stock always has a positive marginal return, and the equilibrium is interior. This is what allows the model to reproduce a stationary regime of heavy litigation rather than its disappearance, and it places us closer to Mailath and Samuelson (2001), where the value of a reputation is bounded by the responsiveness of the short-run players.

2.3 Discounting, delay and judicial interest

Acciarri and Garoupa (2013) show that the effective value of damages depends on the parties' discount rates and opportunity costs, and that "only mere chance will explain the case where both parties have the same discount rate" (p. 37). They emphasise the judicial interest rate:

> *The judicial interest rate not only influences the parties' behaviour during the course of litigation, but it impacts on some of the choices the parties must make prior to litigation, including willingness to commence a lawsuit* (p. 38).

Section 4.1 takes up this channel: a wedge between the employer's and the claimant's discount rates compresses the set of mutually acceptable compositions. We treat that as one force among several rather than as the paper's main mechanism, for a reason worth stating in advance. As Sections 4 and 5 show, the three forces this paper considers do not push in the same direction on the same margin, and presenting them as mutually reinforcing would misdescribe the model.

2.4 Access to justice

The claimant's out-of-pocket exposure is the third institutional object our model uses, and it is the subject of an empirical literature of its own. Yeung and Bruscato (2023) document the incidence and the granting practice of the Brazilian court-fee waiver across a large sample of first-instance cases, and their findings on how routinely the waiver is granted are consistent with what we report in Section 6. What our model adds is that the waiver does not merely improve access: combined with a fee-shifting rule that leaves the losing claimant unexposed and with contingent representation, it produces a lottery whose worst realisation is a gain of zero, and that is what guarantees participation at any degree of risk aversion (Lemma 5.1).

Finally, we note the scarcity of empirical work in this literature, which we attribute to the difficulty of assembling litigation data at scale. Judicial digitisation and text mining have begun to change that, and the exercise in Section 3 is an attempt in that direction.

3 Data and Extraction

The motivation for the model comes from an empirical exercise, and the discipline on the model comes from the same source. This section describes what we collected and how, and states the limitations before reporting anything.

3.1 Source and procedure

We use computational text mining on the open-access public archives of Brazilian labor courts. We downloaded all cases judged between January 2001 and July 2017 by the entry-level courts of the Second Regional Labor Court (TRT-2), which covers the metropolitan area of São Paulo and is the largest labor court in the country. The first retrieval returned 386,995 documents, of which 129,720 are adjudications of labor disputes, comprising sentences, appellate judgments and grievances.

Extraction proceeds in five steps. We index all texts against a keyword list; we extract identifiers such as case number and judge; we parse the structure of each decision into report, legal grounds and dispositive section; we classify the outcome as favourable to the claimant, favourable to the defendant, or dismissal; and we extract monetary values and procedural markers such as fee-waiver requests. Classification relies on regular expressions rather than on supervised learning. This is feasible because Brazilian judgments are highly formulaic: the operative words of a dispositive section are prescribed by procedural law in much the same way that the words of a marriage celebration are, and the small variations that occur are within the reach of a regular expression. We also used cosine similarity to compare and cluster decisions in a separate exercise on forum shopping, which we do not report here.

3.2 Five limitations, stated before any result

Composition rates are not observable. A case resolved by conciliation is closed by a homologation order and produces no reasoned judgment, so such cases are absent from this corpus by construction. This is the binding limitation of the paper, because the composition rate is the observable counterpart of the model's state variable.

The extraction is not yet validated. We report no precision or recall for the outcome classifier. The protocol we are implementing is a manual audit of a random sample of at least one thousand decisions, double-coded, with an inter-coder reliability statistic and a confusion matrix against the automated labels.

The unit of observation is the document, not the case. A single case can generate a sentence, an appellate judgment and one or more grievances. Deduplication by case number is pending, so shares reported below are shares of documents.

Partial victories are pooled with full ones. Brazilian labor claimants routinely cumulate many requests and judges routinely grant some and deny others, so a binary indicator of a claimant victory is close to a mechanical event.[1]

Monetary values are nominal. Amounts are pooled across seventeen years without deflation to a common base year.

3.3 Outcomes

Our sample comprises only cases brought by employees; the much smaller number initiated by employers against employees was excluded.

Table 2: Overall Winning Rates

	Percentage of total
Decisions fully or partially favouring the employee	88.55%
Decisions fully favouring the employer	11.45%

Source: authors' calculations. $n = 129{,}720$ documents. See the caveat on partial victories in Section 3.2.

Read with that caveat, Table 2 delivers one parameter and no more: an estimate of the probability that the employer is ordered to pay a positive amount. It also delivers the observation that organises Section 5.2. An employer that disputes every claim wins only the complement of this rate, near 11.5%, so however aggressively it litigates it cannot accumulate a record of victories. Whatever a reputation does in this market, it cannot be made of winning. Why employers persist under these numbers, and why claimants keep filing, is the question the next two sections address.

4 A Game-Theoretical Model for Labor Litigation

There are costs and benefits associated with litigation, and they are expected rather than deterministic, since a proceeding is lengthy and its outcome uncertain. There are also costs and benefits associated with a composition, in this case more nearly deterministic, because an agreement is a much shorter process. A rational party, claimant or defendant, weighs the two. This section sets out the static framework; Section 5 adds risk aversion and repetition.

[1] We aggregate decisions fully favouring the employee with those partially favouring the employee. It is usual for employees to demand several items at once, such as severance pay, overtime and moral damages, even without expecting to obtain all of them, and there is a corresponding expectation that judges will rarely grant every item. What is unambiguous in the data is the set of cases in which the judge grants nothing. The economically meaningful measure is the ratio of amounts granted to amounts claimed, which requires parsing the initial petition as well as the dispositive section, and which is work in progress.

4.1 The Static Base Model

We consider cases in which the employer owes the employee a sum of money, whether as unpaid benefits or as damages, to be fixed either by judgment or by agreement.

Consider first a game in which the employee decides whether to sue. If dispute is chosen, Nature moves twice: with probability p_c the employee prevails, and conditional on prevailing, with probability p_e the judgment is successfully executed. The parameters are:

- v_s: value of the composition, paid at the moment of agreement;
- v_c: value of the judicial condemnation, paid after successful execution;
- p_c: probability of a condemnation favourable to the employee;
- p_e: probability of successful execution;
- c_w: court costs and expert fees borne by the employee;
- c_e: litigation costs borne by the employer;
- $\phi \in [0, 1)$: the employee's attorney's contingent share of what she recovers.

Remark 4.1 (Attorney fees)**.** *Three features of Brazilian labor procedure fix the fee structure and we build them in from the start rather than carrying general fee parameters that the institutions set to zero. Representation is retained on a* quota litis *basis, so the employee's attorney takes a share ϕ of what she actually recovers and nothing when she recovers nothing; the fee therefore never appears as an outlay and never appears on the employer's side. Under the case law governing our sample period, a losing employee owed the employer no attorney fees on mere sucumbência. And the court-fee waiver sets $c_w = 0$ for the large majority of claimants, as Section 6 documents. We retain c_w and c_e as parameters and set the fee-shifting term to zero throughout.*

The game tree is illustrated in Figure 1.

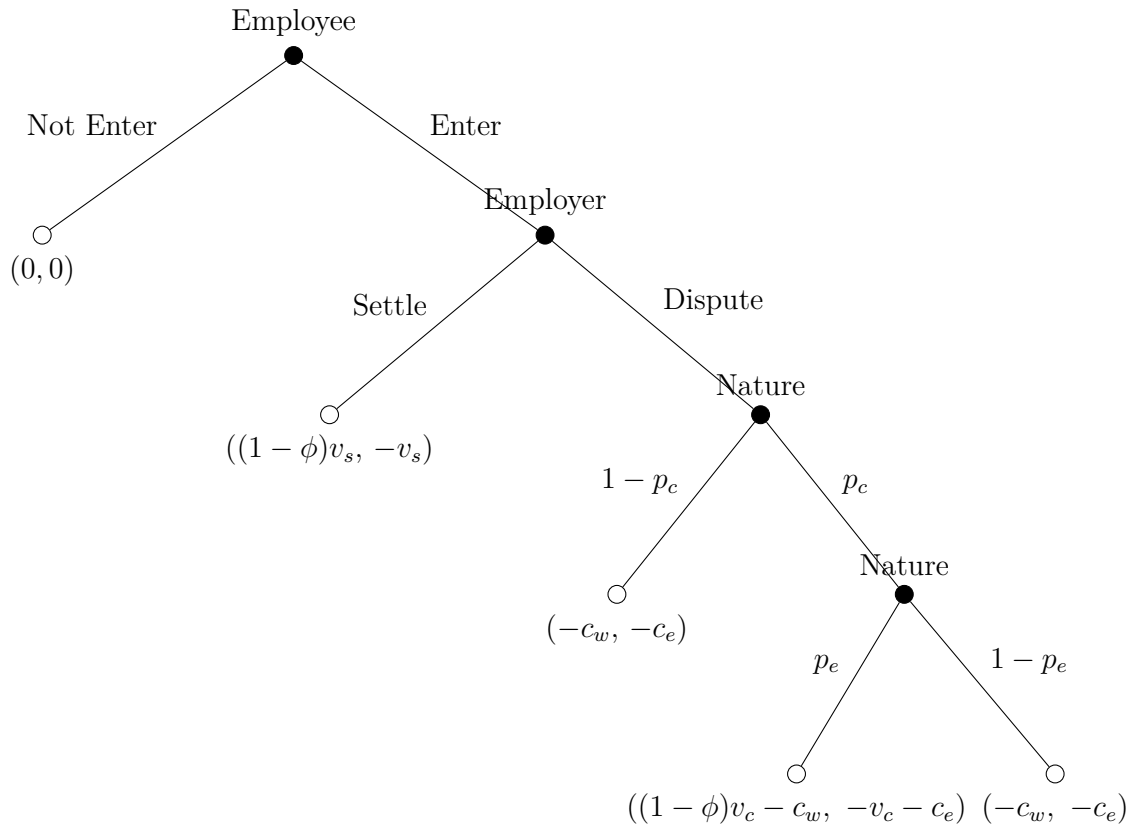


Figure 1: Litigation game tree. The first entry at each terminal node is the employee's payoff, the second the employer's.

Since the employee's attorney is paid out of her recovery, and since the losing employee owes the employer nothing, the loss branch and the failed-execution branch deliver the same payoff pair, $(-c_w,\ -c_e)$. Taking expectations over Nature,

$$\mathbb{E}\Big(u_w(\text{enter, dispute})\Big) = p_c\, p_e\, (1-\phi)\, v_c - c_w, \tag{4.1}$$

$$\mathbb{E}\Big(u_e(\text{enter, dispute})\Big) = -\,p_c\, p_e\, v_c - c_e. \tag{4.2}$$

4.1.1 Employee's decision

The employee prefers to enter if

$$p_c\, p_e\, (1-\phi)\, v_c - c_w > 0 \tag{4.3}$$

and

$$v_s > 0. \tag{4.4}$$

From now on we assume both hold. Note that (4.3) holds automatically when $c_w = 0$, which anticipates Lemma 5.1.

4.1.2 Employer's decision

The employer prefers to dispute rather than to compose if

$$\mathbb{E}\Big(u_e(\text{enter, dispute})\Big) > -v_s \quad\Longleftrightarrow\quad p_c\, p_e\, v_c + c_e < v_s. \tag{4.5}$$

4.1.3 Including the length of trial

Introducing a horizon T and an employer discount rate r_H, condition (4.5) becomes

$$\frac{p_c\, p_e\, v_c + c_e}{(1 + r_H)^T} < v_s. \tag{4.6}$$

Proposition 4.1 (Static equilibrium with time discounting)**.** *Suppose* (4.3) *and* (4.4) *hold. Then the game has a unique subgame perfect equilibrium: the employee enters, and the employer disputes if* (4.6) *holds and composes otherwise.*

A high employer discount rate and a long proceeding both reduce the present-value cost of a condemnation, and therefore make dispute rational even when p_c is high. This is the first of the forces the paper considers, and we develop it next.

4.2 Simultaneous Compositions, Different Discount Rates and a Time Horizon

Suppose the employee still decides whether to enter, but that after entry the two parties simultaneously choose `Settle` or `Dispute`, with a composition requiring the consent of both. If either disputes, the case is adjudicated with the properties already described: it takes T periods, the employee prevails with probability p_c, and execution succeeds with probability p_e.

The trial payoffs are now discounted at different rates, r_H for the employer and r_L for the employee, with $r_H > r_L$. This captures the idea that employers have more opportunities and better information about where to place funds. The game is shown in Figure 2.

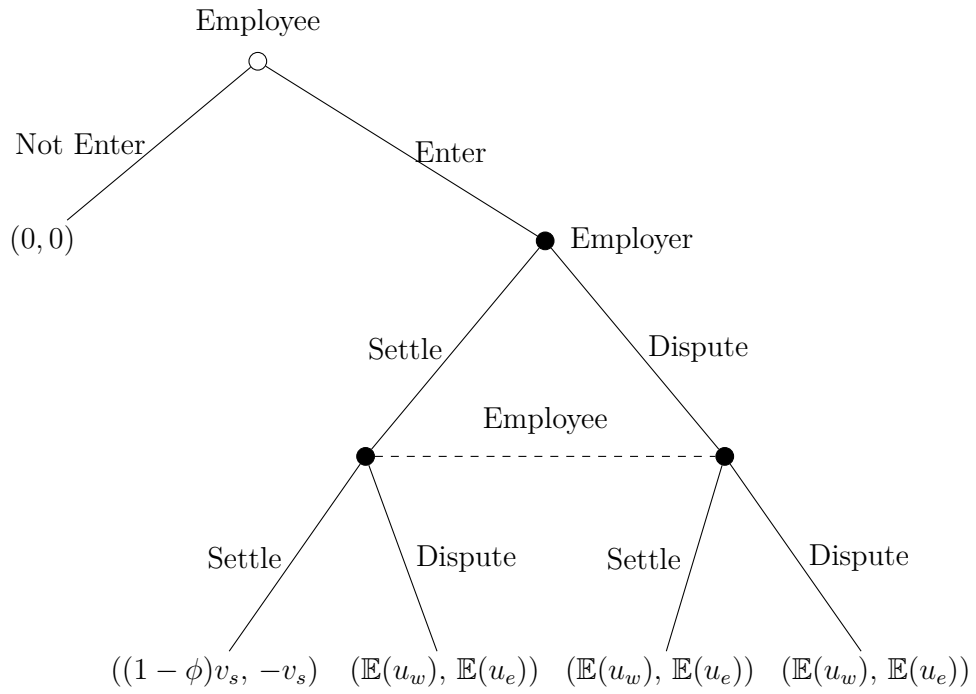


Figure 2: Litigation game with simultaneous composition decisions.

4.2.1 Payoffs and strategies

If Nature is called, the discounted payoffs are

$$\mathbb{E}(u_w) = \frac{p_c\, p_e\, (1-\phi)\, v_c - c_w}{(1+r_L)^T} \tag{4.7}$$

$$\mathbb{E}(u_e) = -\frac{p_c\, p_e\, v_c + c_e}{(1+r_H)^T}. \tag{4.8}$$

Assuming (4.3) and (4.4), the employee enters. She then prefers to dispute if

$$\frac{p_c\, p_e\, (1-\phi)\, v_c - c_w}{(1+r_L)^T} > (1-\phi)\, v_s \tag{4.9}$$

and the employer prefers to dispute if

$$\frac{p_c\, p_e\, v_c + c_e}{(1+r_H)^T} < v_s. \tag{4.10}$$

Remark 4.2 (Why a refinement is needed)**.** *A composition requires the consent of both parties, so a party that expects its opponent to dispute is indifferent between its own two actions: its payoff is the trial payoff either way. Consequently* (`Dispute`, `Dispute`) *is a Nash equilibrium of the simultaneous subgame for* all *parameter values, including those under which both parties strictly prefer to compose. The multiplicity is an artefact of the unanimity requirement, and it is removed by deleting weakly dominated strategies: when* (4.9) *fails,* `Settle` *weakly dominates* `Dispute` *for the employee, and symmetrically for the employer. One round of deletion suffices and each player is left with a unique undominated action, so no question of order arises.*

Proposition 4.2. *Suppose* (4.3) *and* (4.4) *hold. Then the simultaneous subgame has a unique equilibrium in weakly undominated strategies:*

(i) if (4.9) *and* (4.10) *both hold, both parties dispute;*

(ii) if (4.9) *fails and* (4.10) *holds, the employee would compose and the employer disputes, so the case is adjudicated;*

(iii) if (4.9) *holds and* (4.10) *fails, the employer would compose and the employee disputes, so the case is adjudicated;*

(iv) if both fail, both compose and the case is composed.

A composition occurs if and only if (iv) obtains.

Proposition 4.2 exhibits a first asymmetry. The two dispute conditions are discounted at *different* rates over the same horizon, so even when the expected judgment $p_c p_e v_c$ is common to both parties, its present value is not. As r_H rises or T lengthens, the left-hand side of (4.10) shrinks and the employer's dispute condition becomes easier to satisfy: adjudication becomes cheaper in present-value terms while the nominal expected loss is unchanged. The set of compositions acceptable to both parties, that is the set of v_s satisfying neither (4.9) nor (4.10), is compressed as the wedge $r_H - r_L$ widens.[2]

[2]The same sensitivity is why the 2017 labor reform, which introduced procedural rules designed to shorten proceedings, is predicted by this channel to reduce litigation.

Two qualifications should be recorded here rather than left to the reader. The compression operates on the *level* of the acceptable set and says nothing about who files; it is therefore a statement about which cases are adjudicated among those already brought, and not about the flow of claims. And it is not the mechanism this paper ultimately proposes. Section 5.1 shows that claimant risk aversion works in the opposite direction on this same margin, and Section 5.2 shows that a repeat-player defendant will refuse compositions that the one-shot comparison in (4.10) would have it accept. These are alternative accounts of the same margin and they do not point the same way; Section 6.2 asks the data about each of them.

5 Extensions: Risk Aversion and Reputation

The baseline model treats each lawsuit as a one-shot encounter between a worker and an employer, both risk-neutral. We relax the two assumptions in turn. Section 5.1 gives the worker constant relative risk aversion and derives her reservation value exactly. Section 5.2 makes the employer a long-lived player facing a continuing flow of short-lived claimants, whose conduct in the cases of one year is observed by the workers who might file in the next. They are alternatives rather than complements, and Section 6.2 confronts each with the corpus separately.

5.1 Worker Risk Aversion

The base model assumes that both parties evaluate uncertain outcomes by their expected monetary value. This is a natural first approximation for a large corporation, which pools risk across many simultaneous cases. It is systematically violated for an individual worker, for three reasons documented in this setting:

1. *Income loss during litigation*: labor proceedings in Brazil last years, and throughout the dismissed worker faces reduced income or unemployment, so the marginal utility of money is higher than under normal circumstances.

2. *Psychological disutility of uncertainty*: the stress of a prolonged legal dispute is a non-monetary cost that lowers effective utility below any expected-value calculation.

3. *Liquidity constraints*: low-income workers have limited capacity to smooth consumption while waiting, which is precisely what a multi-year proceeding requires of them.

5.1.1 The CRRA Utility Function

Definition 5.1 (CRRA Utility)**.** *The worker's Bernoulli utility function is*

$$U(w) = \begin{cases} \dfrac{w^{1-\rho}}{1-\rho} & \text{if } \rho \neq 1, \quad \rho > 0 \\ \ln(w) & \text{if } \rho = 1 \end{cases} \tag{5.1}$$

where $w > 0$ is terminal wealth and $\rho \geq 0$ is the coefficient of relative risk aversion.

Remark 5.1. *Three canonical cases: $\rho = 0$ gives $U(w) = w$ and recovers the base model; $\rho = 1$ gives logarithmic utility; $\rho > 1$ is moderate to strong risk aversion, which we argue is the empirically relevant range here.*

$U'(w) = w^{-\rho} > 0$ and $U''(w) = -\rho\, w^{-\rho-1} < 0$ for $\rho > 0$, so U is strictly concave, and the Arrow–Pratt coefficient of relative risk aversion is constant at ρ:

$$-\frac{w\, U''(w)}{U'(w)} = \rho. \tag{5.2}$$

5.1.2 Worker's Expected Utility from Litigation

Recall from Remark 4.1 that representation is contingent: the attorney takes a share $\phi \in [0, 1)$ of what the worker recovers, by judgment or by agreement, and nothing when she recovers nothing. A contingent fee scales the worker's recovery but never turns a favourable outcome into a net loss, and that distinction does real work below.

Let $w_0 > 0$ denote initial wealth. Under `Dispute` the worker faces two terminal wealth levels:

- **Win** (probability $p_c\, p_e$): the suit is won and executed, she receives v_c, her attorney takes $\phi\, v_c$, and she bears her own court costs c_w. Terminal wealth $w_0 + (1-\phi)v_c - c_w$.
- **Lose** (probability $1 - p_c\, p_e$): she loses, or the sentence is not executed. No fee is owed and no adverse fee is charged. Terminal wealth $w_0 - c_w$.

Assumption 5.1. *$w_0 > c_w$, so that every wealth level relevant to the worker's decision is strictly positive and $U(\cdot)$ is well defined.*

Her expected utility from entering and disputing is

$$V_E^{\text{dispute}} \;=\; p_c\, p_e\, U\Big(w_0 + (1-\phi)v_c - c_w\Big) \;+\; (1 - p_c\, p_e)\, U\Big(w_0 - c_w\Big), \tag{5.3}$$

from composing at value v_s,

$$V_E^{\text{settle}} \;=\; U\Big(w_0 + (1-\phi)v_s\Big), \tag{5.4}$$

and from staying out,

$$V_E^{\text{out}} \;=\; U(w_0). \tag{5.5}$$

Remark 5.2 (Where the fee enters)**.** *The fee appears* inside *the argument of U, in every state and in both branches, and no monetary quantity is added to or subtracted from a utility index anywhere in what follows. Two consequences. First, the fee is charged on the agreement as well as on the judgment, so it lowers both sides of the worker's dispute-or-compose comparison; under risk neutrality the factor $(1-\phi)$ cancels from that comparison exactly, and under risk aversion it does not, because the certainty equivalent is not homogeneous in the gain when $w_0 > 0$. Second, the residual effect is signed: raising ϕ shrinks the stake and therefore shrinks the risk premium, so a higher contingent share makes disputing* more *attractive relative to composing, not less.*

5.1.3 Certainty Equivalent and Risk Premium

Definition 5.2 (Certainty Equivalent)**.** *The certainty equivalent* CE *is the guaranteed wealth increment that makes the worker indifferent between that increment and disputing:*

$$U(w_0 + \text{CE}) \;=\; p_c\, p_e\, U\Big(w_0 + (1-\phi)v_c - c_w\Big) + (1 - p_c\, p_e)\, U\Big(w_0 - c_w\Big). \tag{5.6}$$

For $\rho \neq 1$,

$$\mathrm{CE} \;=\; \Big[p_c\, p_e \,\big(w_0 + (1-\phi)v_c - c_w\big)^{1-\rho} + (1 - p_c\, p_e)\,\big(w_0 - c_w\big)^{1-\rho}\Big]^{\frac{1}{1-\rho}} - w_0, \tag{5.7}$$

with the logarithmic limit at $\rho = 1$ and $\mathrm{CE} = p_c p_e (1-\phi) v_c - c_w$ at $\rho = 0$.

Lemma 5.1 (Participation)**.** *Suppose $c_w = 0$ and the retainer is purely contingent, so the fee is a share $\phi \in [0, 1)$ of what the worker recovers and is zero when she recovers nothing. Then $\mathrm{CE} > 0$ for every $\rho \geq 0$ and every $w_0 > 0$: filing is individually rational at any degree of risk aversion.*

Proof. Under the stated hypotheses terminal wealth is $w_0 + (1-\phi)v_c$ with probability $p_c p_e > 0$ and w_0 otherwise, so the wealth lottery strictly first-order stochastically dominates the degenerate lottery at w_0. Hence $U(w_0 + \mathrm{CE}) > U(w_0)$, and $\mathrm{CE} > 0$ by strict monotonicity of U. □

Lemma 5.1 is not a technical condition but an institutional fact doing modelling work, and its hypothesis is exactly right. What guarantees participation is that the worker's worst realisation is a gain of zero rather than a net loss, and that requires *both* the waiver of judicial fees, which sets $c_w = 0$, *and* a purely contingent retainer, *and* the fee-shifting rule that leaves her unexposed to the employer's attorney. Any fixed unrecoverable outlay, however small, makes the worst realisation a loss, and then CE turns negative once ρ is large enough and the worker does not file. It is the conjunction of three Brazilian rules, and not any one of them, that produces a lottery with no losing ticket.

Definition 5.3 (Risk Premium)**.**

$$\mathrm{RP} \;\equiv\; \mathbb{E}[\,\mathit{payoff}\,] - \mathrm{CE}, \qquad \mathbb{E}[\,\mathit{payoff}\,] = p_c\, p_e\,(1-\phi)\,v_c - c_w. \tag{5.8}$$

Definition 5.3 is an identity and involves no approximation. The familiar closed form does, and we state it as intuition rather than as a tool.

Remark 5.3 (Second-order approximation, and when it is safe)**.** *For risks small relative to wealth, a second-order expansion of $U(\cdot)$ about $w_0 + \mu$, where $\mu = \mathbb{E}[\,\mathit{payoff}\,]$, gives*

$$\mathrm{RP} \;\approx\; \frac{\rho}{2} \cdot \frac{\varsigma^2}{w_0 + \mu}, \qquad \varsigma^2 = p_c\, p_e\,(1 - p_c\, p_e)\,\Big[(1-\phi)v_c\Big]^2. \tag{5.9}$$

Two cautions. Absolute risk aversion must be evaluated at the mean of terminal wealth, $w_0 + \mu$, and not at w_0; for CRRA the two differ by the factor $(w_0 + \mu)/w_0$, which is large in this application. And the expansion requires ς to be small relative to $w_0 + \mu$, which here it is not: the award is of the same order as, or larger than, a dismissed worker's liquid wealth. Applied outside its range, (5.9) can return a negative certainty equivalent for a lottery whose worst realisation is a gain of zero, which Lemma 5.1 shows is impossible. We therefore use the exact expression (5.7) throughout, and keep (5.9) only for the intuition that the risk premium rises with ρ and with the variance of the stake, and falls with wealth.

5.1.4 Actions and Payoffs under Risk Aversion

We embed CRRA preferences in the game of Section 4.2. The structure is unchanged.

Worker's payoffs (in utility terms):

$$V_E^{\text{dispute}} = p_c\, p_e\, U\Big(w_0 + (1-\phi)v_c - c_w\Big) + (1 - p_c\, p_e)\, U\Big(w_0 - c_w\Big) \tag{5.10}$$

$$V_E^{\text{settle}} = U\Big(w_0 + (1-\phi)v_s\Big) \tag{5.11}$$

$$V_E^{\text{out}} = U(w_0) \tag{5.12}$$

Employer's payoffs (monetary, risk-neutral):

$$V_F^{\text{dispute}} = -\frac{p_c\, p_e\, v_c + c_e}{(1+r_H)^T} \tag{5.13}$$

$$V_F^{\text{settle}} = -v_s \tag{5.14}$$

5.1.5 Participation and Dispute Conditions

The worker **enters** if and only if

$$\max\Big\{V_E^{\text{dispute}},\ V_E^{\text{settle}}\Big\} > U(w_0), \tag{PC-RA}$$

which by Lemma 5.1 holds automatically whenever $c_w = 0$. She **disputes** rather than composes if and only if $V_E^{\text{dispute}} > V_E^{\text{settle}}$, that is, by Definition 5.2 and strict monotonicity of U,

$$\text{CE} > (1-\phi)\, v_s. \tag{DC-RA}$$

Equivalently, in monetary terms and using $\text{CE} = \mathbb{E}[\,\text{payoff}\,] - \text{RP}$,

$$p_c\, p_e\, (1-\phi)\, v_c - c_w - \text{RP} > (1-\phi)\, v_s. \tag{DC-RA$'$}$$

The employer disputes if and only if

$$\frac{p_c\, p_e\, v_c + c_e}{(1+r_H)^T} < v_s. \tag{DC-F}$$

Conditions (DC-RA) and (DC-RA$'$) are exact and equivalent, and no monetary quantity is subtracted from a utility index in either.

5.1.6 Equilibrium under Risk Aversion

Proposition 5.1 (Equilibrium with worker risk aversion)**.** *Suppose Assumption 5.1 holds, the worker has CRRA preferences with $\rho \geq 0$, and (PC-RA) holds so that she enters. Then, by the argument of Remark 4.2, the simultaneous subgame has a unique equilibrium in weakly undominated strategies:*

(i) ***Both dispute*** *if and only if (DC-RA) and (DC-F) both hold.*

(ii) ***Worker composes, employer disputes*** *if and only if (DC-RA) fails and (DC-F) holds.*

(iii) ***Worker disputes, employer composes*** *if and only if* (DC-RA) *holds and* (DC-F) *fails.*

(iv) ***Both compose*** *if and only if both fail.*

The case is composed if and only if (iv) obtains.

Proof. Consider the worker. Playing `Settle` yields V_E^{settle} when the employer composes and V_E^{dispute} when it disputes; playing `Dispute` yields V_E^{dispute} against either action. Hence if $V_E^{\text{settle}} > V_E^{\text{dispute}}$ then `Settle` weakly dominates `Dispute`, and if the inequality is reversed then `Dispute` weakly dominates `Settle`; the comparison is (DC-RA) by Definition 5.2. The argument for the employer is identical with (DC-F) in place of (DC-RA). Deleting the weakly dominated action for each player leaves a unique action profile in each of the four exhaustive and mutually exclusive configurations. A composition requires unanimity, hence case (iv). The entry node follows by backward induction, since the worker's continuation value is $\max\{V_E^{\text{dispute}}, V_E^{\text{settle}}\} > U(w_0)$ by (PC-RA). □

Corollary 5.1 (Risk aversion expands the composition region)**.** *Fix* $(p_c, p_e, v_c, c_w, \phi, v_s, T, r_H)$. CE *is strictly decreasing in* ρ *and strictly increasing in* w_0*, so increasing* ρ *or decreasing* w_0 *makes* (DC-RA) *more likely to fail and weakly expands the set of composition offers the worker accepts.*

Proof. By (DC-RA) the worker disputes if and only if $\text{CE} > (1-\phi)v_s$, so it suffices to sign the dependence of CE on (ρ, w_0). The wealth lottery in (5.6) does not depend on ρ, so comparing two CRRA utilities with $\rho' > \rho$ compares a more concave with a less concave transformation of the same lottery, and Pratt's theorem gives $\text{CE}(\rho') < \text{CE}(\rho)$; equivalently, the bracketed expression in (5.7) is the $(1-\rho)$-power mean of the two terminal wealth levels and is strictly decreasing in ρ. CRRA exhibits decreasing absolute risk aversion, so CE is strictly increasing in w_0. Both statements are exact and use no approximation. □

Corollary 5.1 runs against the intuition that risk aversion should explain the absence of compositions. It does the opposite: a risk-averse worker demands *less* to compose, which makes agreement easier, not harder. Risk aversion is therefore not a mechanism that suppresses composition, and this is the reason Section 6.2.1 concludes that it cannot on its own account for the phenomenon this paper studies. Section 5.2 sets out a different mechanism, one that does not require the claimant to be risk averse.

5.2 Employer Reputation in a Repeated Game

The previous subsections treat each lawsuit in isolation. This misrepresents the position of a large Brazilian employer, which faces a continuing flow of claims and whose conduct in the cases of this year is observed by the workers who might file next year. We now model the employer as a long-lived player facing a sequence of short-lived claimants.

One convention is refined here. Sections 4 and 5.1 treat a composition as paid at the moment of agreement. We now date that moment at t_s months after filing, when the conciliation hearing takes place, against T months for a judgment, with $1 \le t_s < T$. Every object in this section is measured at the date of filing. Write

$$v_s^W \equiv (1-\phi)\, v_s\, (1+r_L)^{-t_s}, \qquad v_s^F \equiv v_s\, (1+r_H)^{-t_s}, \qquad \widehat{\text{CE}} \equiv \text{CE}\, (1+r_L)^{-T} \tag{5.15}$$

for the value of a composition to the worker and to the employer, and for the certainty equivalent of the judgment branch, all three at the date of filing. The third of these

corrects a convention. In Sections 4 and 5.1 the two branches were resolved at a common date, so no discount factor distinguished them. Here they are not: a judgment is collected at T and a composition at t_s, so the judgment branch must carry $(1+r_L)^{-T}$ wherever it is set against v_s^W. Since $\widehat{\text{CE}} < \text{CE}$, the correction works in favour of the mechanism, and Lemma 5.4 shows exactly how.

5.2.1 Structure of the repeated game

Time is discrete, $t = 0, 1, 2, \ldots$, and one period is one month. The firm is infinitely lived and discounts across periods at $\delta = 1/(1+r_H) \in (0,1)$. In each period a cohort of mass M of potential claimants arrives; each lives one period and files at most one claim. Within period t:

1. Each potential claimant i draws a *participation cost* $\varepsilon_i \geq 0$, borne in every month her claim remains alive.
2. The firm's public record $r_t \in [0,1]$ is observed.
3. Each claimant chooses `File` or `Drop`.
4. The firm chooses $q_t \in [0,1]$, the share of filed claims it takes to judgment, composing the remainder at the going value v_s.
5. Disputed claims are adjudicated and executed at T; composed claims are paid at t_s.
6. The record is updated.

Assumption 5.2 (Heterogeneous participation costs)**.** *The ε_i are drawn independently from a distribution G with continuous, strictly positive density g on $[0, \bar{\varepsilon}]$. They are the only source of heterogeneity among claimants.*

The participation cost is a money-metric measure of the time, the stress, the foregone opportunities and the perceived risk of being marked as litigious that are involved in keeping a claim alive. It is not an outlay: it does not reduce terminal wealth in any state and does not enter the argument of $U(\cdot)$, which is what keeps Lemma 5.1 intact. Writing

$$A(\tau) \;\equiv\; \sum_{k=1}^{\tau} (1+r_L)^{-k} \;=\; \frac{1-(1+r_L)^{-\tau}}{r_L} \tag{5.16}$$

for the present value of one unit of that cost borne in each of τ months, a claim carried to judgment costs her $\varepsilon_i A(T)$ and a claim composed at the hearing costs her $\varepsilon_i A(t_s)$.

5.2.2 Reputation as a record of disputing

Definition 5.4 (Firm reputation)**.** *The firm's reputation $r_t \in [0,1]$ is the exponentially smoothed share of past claims it took to judgment rather than composing:*

$$r_{t+1} \;=\; (1-\lambda)\, r_t \;+\; \lambda\, q_t, \qquad \lambda \in (0,1], \tag{5.17}$$

where λ is the speed at which the labour pool's information turns over and $1/\lambda$ is the effective memory of the market.

Remark 5.4 (Why toughness and not victory)**.** *Reputation is defined over the firm's composition policy and not over its record of winning. Two reasons. First, the firm controls the former and not the latter: under a policy of disputing everything its win rate converges to* $1 - p_c$*, which Table 2 puts near* 0.115*, so a firm that fights every claim still loses most of them and a reputation made of victories is one this firm cannot build. Second, what a prospective claimant learns from her union, her lawyer and her former colleagues is whether a given employer composes, which is also the object our firm-level data record. The mechanism is then the one Galanter (1974) describes: what deters a claimant is not the belief that she will lose, but the near-certainty that she will wait.*

5.2.3 Entry with heterogeneous claimants

A claimant facing a firm with record r expects her claim to be adjudicated with probability r and composed with probability $1 - r$. Her value from the judgment branch is $\widehat{\mathrm{CE}}$, the dated certainty equivalent of Definition 5.2, and from the composition branch v_s^W.

Assumption 5.3 (Linear evaluation of the adjudication risk)**.** *At the filing stage the claimant values the composite prospect linearly in* r*: facing a firm with record* r *is worth* $r\,\widehat{\mathrm{CE}} + (1 - r)\,v_s^W$ *to her.*

She files if and only if

$$r\,\widehat{\mathrm{CE}} + (1 - r)\,v_s^W \;>\; \varepsilon_i \Big[\, r\,A(T) + (1 - r)\,A(t_s) \,\Big]. \tag{5.18}$$

Both sides are dated at filing, and the participation-cost stream is discounted at r_L like everything else on the claimant's side.

Remark 5.5 (What Assumption 5.3 buys and what it costs)**.** *Section 5.1 makes the claimant risk averse over the* outcome of adjudication*, which is where the risk to her wealth is: she recovers* $(1 - \phi)v_c$ *or she recovers nothing. Assumption 5.3 says that she does not price a second layer of risk over* which *of the two known prospects she will face. Two things recommend it. The uncertainty it neutralises is resolved early, at* t_s*, and it is not a gamble over her wealth but over which branch she enters; and* r *is a frequency observed in a cohort rather than a subjective probability, so reading it without a risk premium of its own is the natural counterpart of the interpretation in Remark 5.4.*

It is not innocuous, and the cost should be stated. If instead the claimant applied CRRA to the compound prospect, the numerator of ε^* *below would be the certainty equivalent of a three-outcome lottery rather than a linear mixture, and* ε^* *would in general fail to be monotone in* r*: for large* ρ *it can fall over an initial range of* r *and rise thereafter, which would reverse the sign of the entry response to the record and remove the mechanism. Lemma 5.2 is therefore a statement about the entry rule of Assumption 5.3, and not a claim that every risk-averse claimant is deterred by a record of toughness. Relaxing this is the most obvious extension of the model.*

Definition 5.5 (Entry threshold and entry mass)**.**

$$\varepsilon^*(r) \;\equiv\; \frac{r\,\widehat{\mathrm{CE}} + (1 - r)\,v_s^W}{r\,A(T) + (1 - r)\,A(t_s)}, \qquad n(r) \;=\; M\,G\Big(\varepsilon^*(r)\Big). \tag{5.19}$$

Claimant i *files if and only if* $\varepsilon_i < \varepsilon^*(r)$*.*

Lemma 5.2 (Deterrence condition)**.** *ε^* is strictly decreasing on $[0,1]$, and hence $n'(r) < 0$, if and only if*

$$\widehat{\mathrm{CE}}\,A(t_s) \;<\; v_s^W\,A(T), \qquad \textit{equivalently} \qquad \frac{\widehat{\mathrm{CE}}}{v_s^W} \;<\; \frac{A(T)}{A(t_s)}. \tag{5.20}$$

The sign of $\varepsilon^{*\prime}$ does not depend on r.

Proof. Write $N(r) = r\widehat{\mathrm{CE}} + (1-r)v_s^W$ and $E(r) = rA(T) + (1-r)A(t_s)$, so $\varepsilon^* = N/E$ with $N' = \widehat{\mathrm{CE}} - v_s^W$ and $E' = A(T) - A(t_s)$. Then

$$\varepsilon^{*\prime}(r) = \frac{N'E - NE'}{E^2} = \frac{(\widehat{\mathrm{CE}} - v_s^W)\big[A(t_s) + r\big(A(T) - A(t_s)\big)\big] - \big[v_s^W + r(\widehat{\mathrm{CE}} - v_s^W)\big]\big(A(T) - A(t_s)\big)}{E^2}.$$

The two terms in r are $r(\widehat{\mathrm{CE}} - v_s^W)(A(T) - A(t_s))$ with opposite signs and cancel identically, leaving

$$\varepsilon^{*\prime}(r) = \frac{(\widehat{\mathrm{CE}} - v_s^W)A(t_s) - v_s^W\big(A(T) - A(t_s)\big)}{E^2} = \frac{\widehat{\mathrm{CE}}\,A(t_s) - v_s^W\,A(T)}{E^2},$$

whose sign is that of $\widehat{\mathrm{CE}}A(t_s) - v_s^W A(T)$, independent of r. □

A reputation for toughness deters entry if and only if the ratio of the two *values* is smaller than the ratio of the two *delays*, each measured as a discounted annuity. Two observations bound the content of that condition.

Remark 5.6 (The mechanism runs on delay, not on merits)**.** *If claimants were patient and bore no participation cost, the right-hand side of (5.18) would vanish and a claimant would file whenever $r\widehat{\mathrm{CE}} + (1-r)v_s^W > 0$, which is increasing in r whenever $\widehat{\mathrm{CE}} > v_s^W$. A tough firm would then* attract *claims rather than deter them, for the straightforward reason that a judgment pays more than a composition. What makes toughness a deterrent is not the outcome it delivers but the time it takes to deliver it. The mechanism therefore requires claimants who cannot costlessly wait, and we treat that as an object to be disciplined by the data rather than as a free normalisation. A claimant's risk aversion lowers $\widehat{\mathrm{CE}}$ and so relaxes (5.20), which is the channel by which Corollary 5.1 would bear on this section; we do not pursue it, for the reason given in Section 6.2.1.*

Remark 5.7 (The condition is slack when compositions price at the reservation value)**.** *If the firm composes at the claimant's reservation value, so that $v_s^W = \widehat{\mathrm{CE}}$, then $\widehat{\mathrm{CE}}/v_s^W = 1 < A(T)/A(t_s)$ and (5.20) holds automatically, with deterrence operating through delay alone. The condition can fail only when compositions are priced above the claimant's reservation value by a factor exceeding $A(T)/A(t_s)$, which is not a configuration we regard as relevant.*

Lemma 5.3 (No absorbing state)**.** *Under Assumption 5.2, $c_w = 0$ and a purely contingent retainer, $\varepsilon^*(r) > 0$ for every $r \in [0,1]$, and therefore*

$$n(r) \;\geq\; M\,G\!\left(\min\left\{\frac{\widehat{\mathrm{CE}}}{A(T)},\; \frac{v_s^W}{A(t_s)}\right\}\right) \;>\; 0.$$

Litigation never ceases, at any reputation the firm can attain.

Proof. $\widehat{\mathrm{CE}} > 0$ for every $\rho \geq 0$ by Lemma 5.1, and $v_s^W \geq 0$, so the numerator of ε^* in (5.19) is strictly positive on $[0,1]$. Since ε^* is monotone by Lemma 5.2, its minimum over $[0,1]$ is attained at an endpoint, where it equals $\widehat{\mathrm{CE}}/A(T)$ or $v_s^W/A(t_s)$. The density of G is strictly positive at every interior point of its support. □

Lemma 5.3 is the substantive gain from allowing claimants to differ. Under a homogeneous population, entry is an all-or-nothing decision, so successful deterrence is an absorbing state in which filings stop forever, which is the opposite of the phenomenon this paper sets out to explain. With heterogeneous participation costs, deterrence is always *partial*: however tough the firm becomes, the claimants with the lowest personal cost still sue, the reputational stock retains a strictly positive marginal return, and the stationary outcome is persistent litigation at a reduced rate.

5.2.4 The firm's intertemporal problem

Let

$$c \;\equiv\; \frac{p_c\, p_e\, v_c + c_e}{(1+r_H)^T} \tag{5.21}$$

denote the present value to the firm of taking one claim to judgment, against v_s^F from (5.15) for composing one. The firm solves

$$W(r) \;=\; \max_{q\in[0,1]} \Big\{ -n(r)\big[q\,c + (1-q)\,v_s^F\big] \;+\; \delta\, W\Big((1-\lambda)r + \lambda q\Big) \Big\}. \tag{5.22}$$

Two objects summarise the trade-off:

$$\kappa \;\equiv\; 1 - \frac{v_s^F}{c} \qquad \text{and} \qquad \eta(r) \;\equiv\; -\frac{n'(r)}{n(r)} \;>\; 0. \tag{5.23}$$

Assumption 5.4 (Static cost of toughness)**.** *$0 < v_s^F < c$, that is $\kappa \in (0,1)$.*

If Assumption 5.4 failed, composing would be more expensive than adjudicating, the firm would refuse even in a one-shot encounter, and there would be nothing for a reputation to explain.

Remark 5.8 (Semi-elasticity, not elasticity)**.** *The object that enters the first-order condition is $-n'/n$ and not the ordinary elasticity $-r\,n'/n$. Reputation is a share and not a quantity in natural units, so the economically meaningful magnitude is the proportional fall in entry per unit of r. The distinction is substantive at $r = 0$, where the ordinary elasticity is identically zero and would deliver the false conclusion that permanent accommodation is always optimal.*

5.2.5 Stationary equilibrium

Definition 5.6 (Stationary equilibrium)**.** *A stationary equilibrium is a pair $(\bar r, \bar q)$ such that $\bar q$ attains the maximum in (5.22) at $r = \bar r$, and $\bar r = \bar q$. In a stationary equilibrium the belief that a claim is adjudicated with probability $\bar r$ is correct, since the firm in fact adjudicates a share $\bar q = \bar r$ of the cohort.*

We restrict attention to stationary equilibria, and state what the restriction buys and what it costs. It buys correctness of claimants' beliefs by construction, so no bounded rationality is assumed anywhere. It costs the analysis of transitions, along which a record summarising past conduct is not a sufficient statistic for present conduct.

Proposition 5.2 (Stationary equilibrium and the patience threshold)**.** *Suppose Assumptions 5.2 and 5.4 hold and the deterrence condition (5.20) is satisfied. Then:*

(i) In any stationary equilibrium $\bar{r} = \bar{q}$, so the state of the model is observable as one minus the firm's composition rate.

(ii) Permanent toughness, $\bar{q} = 1$, requires

$$\delta \;\geq\; \delta^* \;\equiv\; \frac{\kappa}{\lambda\,\eta(1) + \kappa\,(1-\lambda)}\,: \tag{5.24}$$

if $\delta < \delta^*$ the firm strictly gains by composing a positive share of the cohort. The condition is necessary; on its sufficiency see Remark 5.9.

(iii) $\delta^* < 1$ if and only if $\eta(1) > \kappa$: the marginal deterrence delivered by the record must exceed the static cost of producing it.

(iv) Permanent accommodation, $\bar{q} = 0$, requires $\delta \leq \delta_0 \equiv \kappa/\big[\lambda\,\eta(0)(1-\kappa) + \kappa(1-\lambda)\big]$. If in addition

$$\eta(0)\,(1-\kappa) \;>\; \eta(1), \tag{5.25}$$

then $\delta_0 < \delta^*$ and the two corner conditions partition the patience line, so that for $\delta_0 < \delta < \delta^*$ the stationary dispute rate is interior. By Lemma 5.3 entry is strictly positive in every case.

Proof. Part (i) is immediate from (5.17): $r_{t+1} = r_t$ requires $r_t = q_t$.

For (ii), write $\Psi(q; r) = -n(r)[qc + (1-q)v_s^F] + \delta W((1-\lambda)r + \lambda q)$. Then

$$\frac{\partial \Psi}{\partial q} = -\,n(r)(c - v_s^F) + \delta\lambda\,W'\Big((1-\lambda)r + \lambda q\Big).$$

By Assumption 5.4 the first term is strictly negative, so any interior or upper-corner optimum requires $W' > 0$. The envelope condition is

$$W'(r) = -\,n'(r)\Big[qc + (1-q)v_s^F\Big] + \delta(1-\lambda)W'(r'),$$

and $-n'(r) > 0$ by Lemma 5.2, confirming $W' > 0$. At the candidate $r = q = 1$, where $r' = 1$,

$$W'(1) = \frac{-\,n'(1)\,c}{1 - \delta(1-\lambda)},$$

and $q = 1$ is optimal if and only if $\partial\Psi/\partial q \geq 0$ there, that is $\delta\lambda(-n'(1))c/[1-\delta(1-\lambda)] \geq n(1)(c - v_s^F)$. Dividing by $n(1)c > 0$ and using $\eta(1) = -n'(1)/n(1)$ and $\kappa = (c - v_s^F)/c$ gives $\delta\lambda\eta(1)/[1-\delta(1-\lambda)] \geq \kappa$, which rearranges to $\delta[\lambda\eta(1) + \kappa(1-\lambda)] \geq \kappa$ and hence to (5.24).

For (iii), $\delta^* < 1$ if and only if $\kappa < \lambda\eta(1) + \kappa - \kappa\lambda$, that is $\eta(1) > \kappa$.

For (iv), evaluate the same conditions at $r = q = 0$, where $W'(0) = -n'(0)v_s^F/[1-\delta(1-\lambda)]$ and $q = 0$ is optimal if and only if $\delta\lambda W'(0) \leq n(0)(c - v_s^F)$. Dividing by $n(0)c$ and using $v_s^F/c = 1 - \kappa$ gives the stated bound. Neither corner is optimal for $\delta \in (\delta_0, \delta^*)$, and Ψ is continuous in q on the compact set $[0, 1]$, so a maximiser exists and lies in the interior. Strict positivity of entry is Lemma 5.3. □

Remark 5.9 (What is and is not established)**.** *The conditions in Proposition 5.2 are the first-order and envelope conditions evaluated at the candidate policies. We do not prove concavity of W. Without it these conditions are necessary and not sufficient: $\delta \geq \delta^*$ does not by itself establish that $q = 1$ maximises the right-hand side of (5.22) globally, and we do not claim uniqueness of the stationary equilibrium. The gap is not merely formal. In the numerical example of Remark 5.10 the smallest δ at which $q = 1$ is in fact the global maximiser is $\delta \approx 0.550$, against $\delta^* \approx 0.432$. Any statement that needs a global comparison, as Corollary 5.4 does, therefore carries that comparison explicitly instead of invoking Proposition 5.2(ii).*

Remark 5.10 (Why (5.25) has to be assumed)**.** *Comparing the two thresholds, $\delta_0 < \delta^*$ if and only if $\lambda\big[\eta(0)(1-\kappa) - \eta(1)\big] > 0$, which is (5.25). Nothing in Assumptions 5.2 and 5.4 delivers it, and it can fail. Take G uniform, $A(t_s) = 1$, $A(T) = 10$, $\widehat{\mathrm{CE}} = 1$, $v_s^W = 3$, $c = 1$, $v_s^F = 0.2$ and $\lambda = 1/2$. Then $\kappa = 0.8$, $\eta(1) = 2.9$ and $\eta(0) = 29/3$, so $\eta(0)(1-\kappa) = 1.93 < 2.9 = \eta(1)$, and $\delta^* \approx 0.432$ while $\delta_0 \approx 0.585$. The interior interval is empty and the ordering of defendants into three regimes has no content. Substantively, (5.25) says that entry responds much more sharply to the record at $r = 0$ than at $r = 1$, relative to the static cost of toughness, which is what makes an intermediate record worth holding.*

Corollary 5.2 (Comparative statics of the patience threshold)**.** *Whenever $\eta(1) > \kappa$:*

(a) $\partial\delta^*/\partial\eta(1) < 0$: more responsive entry makes toughness easier to sustain.

(b) $\partial\delta^*/\partial\lambda < 0$: a market with shorter memory, in which the record forms faster, makes toughness easier to sustain.

(c) $\partial\delta^*/\partial\kappa > 0$: a larger static cost of toughness makes it harder to sustain.

Proof. Write $\delta^* = \kappa/[\kappa + \lambda(\eta(1) - \kappa)]$. Part (a) is immediate. For (b), the denominator is increasing in λ when $\eta(1) > \kappa$. For (c), $\delta^* = 1/[1 + \lambda\eta(1)/\kappa - \lambda]$ is increasing in κ. □

Part (c) connects this subsection to Section 4.1. Since $v_s^F/c = \big[v_s/(p_c p_e v_c + c_e)\big](1 + r_H)^{T-t_s}$, a higher employer discount rate r_H raises v_s^F/c, lowers κ and lowers δ^*; the same is true of a lower marginal litigation cost c_e, which is the empirically relevant case for employers maintaining permanent in-house legal departments. Whatever makes delay cheap for the firm also makes a reputation for toughness cheap to acquire.

Corollary 5.3 (Reputation as a partial barrier to entry)**.** *In a stationary equilibrium with $\bar{q} = 1$ the firm disputes claims whose one-shot payoff favours composition, and entry falls from $n(0) = M\,G\big(v_s^W/A(t_s)\big)$ to $n(1) = M\,G\big(\widehat{\mathrm{CE}}/A(T)\big)$, a strict reduction under (5.20). It does not fall to zero. The firm buys a permanent reduction in the flow of claims against it, not the elimination of that flow, and pays for it in every period.*

5.2.6 The deterrence elasticity in closed form

The threshold δ^* is stated in terms of $\eta(1)$, which is a property of the entry margin rather than a primitive. It has a closed form.

Definition 5.7 (Local elasticity of the participation-cost distribution)**.** *At any entry threshold ε^* in the interior of the support of G, let*

$$\alpha \;\equiv\; \frac{\varepsilon^*\, g(\varepsilon^*)}{G(\varepsilon^*)} \;>\; 0 \tag{5.26}$$

denote the local elasticity of G at the margin. For G uniform on $[0, \bar{\varepsilon}]$, $\alpha = 1$.

Lemma 5.4 (Deterrence in closed form)**.** *Evaluated at $r = 1$,*

$$\eta(1) \;=\; \alpha \left[\frac{v_s^W}{\widehat{\mathrm{CE}}} \;-\; \frac{A(t_s)}{A(T)} \right], \tag{5.27}$$

and $\eta(1) > 0$ if and only if the deterrence condition (5.20) holds.

Proof. By Definition 5.5, $n(r) = M\, G(\varepsilon^*(r))$, so $\eta(r) = -n'(r)/n(r) = -\varepsilon^{*\prime}(r)\, g(\varepsilon^*)/G(\varepsilon^*)$. At $r = 1$, $\varepsilon^*(1) = \widehat{\mathrm{CE}}/A(T)$ and, by the proof of Lemma 5.2, $\varepsilon^{*\prime}(1) = [\widehat{\mathrm{CE}} A(t_s) - v_s^W A(T)]/A(T)^2$. Substituting and writing $g/G = \alpha/\varepsilon^*(1)$,

$$\eta(1) = \alpha\, \frac{-\varepsilon^{*\prime}(1)}{\varepsilon^*(1)} = \alpha\, \frac{v_s^W A(T) - \widehat{\mathrm{CE}} A(t_s)}{A(T)^2} \cdot \frac{A(T)}{\widehat{\mathrm{CE}}} = \alpha \left[\frac{v_s^W}{\widehat{\mathrm{CE}}} - \frac{A(t_s)}{A(T)} \right].$$

The sign statement is immediate, since $v_s^W/\widehat{\mathrm{CE}} > A(t_s)/A(T)$ is (5.20). □

Two readings of (5.27) are worth recording, and a third is deliberately withheld. The record is more productive the cheaper a composition looks beside a judgment, $v_s^W/\widehat{\mathrm{CE}}$ large, and the longer adjudication takes beside conciliation, $A(t_s)/A(T)$ small. Nothing in this section requires the claimant to be risk averse: setting $\rho = 0$ leaves $\widehat{\mathrm{CE}} = [p_c p_e (1-\phi) v_c - c_w](1 + r_L)^{-T}$ and every result of Section 5.2 stands. Whether the claimant's risk aversion, which lowers $\widehat{\mathrm{CE}}$ and therefore raises $\eta(1)$, is quantitatively important for δ^* is a question this paper does not answer; we return to it in Section 6.4.

Remark 5.11 (What α is, and why a corpus of decisions cannot see it)**.** *α is a local property of G at the entry margin, so it governs how many* additional *claimants a marginal change in the record deters. It is the one object in the model that a corpus of* decided *cases cannot discipline at all, for a reason that is not a defect of our extraction: such a corpus contains no information whatever about people who considered suing and did not. Any quantitative statement about δ^* will therefore rest on an assumption about G, and should be reported as such.*

5.2.7 Toughness against permanent accommodation

Proposition 5.2(ii) is a local condition, and Remark 5.9 explains why it cannot by itself establish that permanent toughness is worth more than the alternatives. The comparison against the leading alternative, composing every claim from now on, can be made globally and in closed form.

Definition 5.8 (Deterrence dividend)**.** *Let*

$$D(\delta, \lambda) \;\equiv\; (1-\delta) \sum_{t \geq 0} \delta^t\, \frac{n\big((1-\lambda)^t\big)}{n(1)} \tag{5.28}$$

be the discounted average factor by which the filing rate exceeds its toughness level $n(1)$ along the path the record follows when a firm at $r = 1$ composes every claim from then on. Under (5.20), $D(\delta, \lambda) \geq 1$, and D is strictly increasing in δ.

Lemma 5.5 (Toughness against permanent accommodation)**.** *Starting from $r = 1$, disputing every claim forever is worth more to the firm than composing every claim forever if and only if*

$$D(\delta, \lambda) \;>\; \frac{1}{1-\kappa}. \tag{5.29}$$

Proof. Disputing every claim holds the record at $r = 1$ and costs $n(1)\,c$ in every period, so its value is $-n(1)c/(1-\delta)$. Composing every claim drives the record along $r_t = (1-\lambda)^t$ and costs $n(r_t)\,v_s^F$ in period t, so its value is $-v_s^F \sum_{t\geq 0} \delta^t n\Big((1-\lambda)^t\Big)$. The first exceeds the second if and only if $n(1)c/(1-\delta) < v_s^F \sum_{t\geq 0} \delta^t n\Big((1-\lambda)^t\Big)$. Divide by $n(1)c$ and use $v_s^F = (1-\kappa)c$ from Assumption 5.4 to obtain $1/(1-\delta) < (1-\kappa) \sum_{t\geq 0} \delta^t n\Big((1-\lambda)^t\Big)/n(1)$, which is (5.29). For the two statements in Definition 5.8: under (5.20) n is decreasing and $(1-\lambda)^t$ falls in t, so $n\Big((1-\lambda)^t\Big)/n(1)$ is non-decreasing in t and at least one at $t = 0$; a discounted average of a non-decreasing sequence is at least its first term and increases as δ shifts weight towards later terms. □

Corollary 5.4 (Win-win litigation equilibrium)**.** *Suppose the deterrence condition (5.20) holds, $\delta \geq \delta^*$, and the global condition (5.29) holds. Then in the stationary equilibrium with $\bar{q} = \bar{r} = 1$:*

(i) ***Every claimant who files strictly prefers to file.*** *A claimant with $\varepsilon_i < \varepsilon^*(1) = \widehat{\mathrm{CE}}/A(T)$ obtains $\widehat{\mathrm{CE}} - \varepsilon_i A(T) > 0$ against an outside option of zero, and by Lemma 5.1 the set of such claimants is non-empty at every ρ.*

(ii) ***The employer prefers disputing every claim to composing every claim.*** *By Lemma 5.5, $W(1)$ under $\bar{q} = 1$ strictly exceeds the value of composing every claim forever.*

(iii) ***Litigation persists.*** *By Lemma 5.3 the filing rate is strictly positive and constant in every period.*

Condition (5.29) is not implied by $\delta \geq \delta^*$, and the distinction matters. In the example of Remark 5.10 at $\delta = 0.50$, which satisfies $\delta \geq \delta^* \approx 0.432$ with room to spare, one has $D = 4.54$ against $1/(1-\kappa) = 5$: composing every claim forever is worth -0.061 and disputing every claim forever is worth -0.067, so the employer's side of the "win" is absent. Condition (5.29) first holds at $\delta \approx 0.528$. Both conditions are needed, and both are threshold conditions in patience.

The sense in which this is a "win-win" should be stated precisely, because it is not the sense in which a bargain is one. Each party is doing strictly better than its own alternative: the claimant better than not filing, the employer better than composing every claim forever. Both are playing a best response to the other, and neither is being exploited or behaving irrationally. What is *not* true is that the outcome is jointly efficient. In every period the employer pays more than a composition would have cost it, and the claimants who do file wait T months rather than t_s. Nor does the label extend to the claimants the record deters, whose participation cost exceeds $\varepsilon^*(1)$ and who abandon a claim they would have brought against a softer defendant; they are outside the accounting on both sides. The equilibrium is privately rational for those who participate and jointly wasteful, and the two statements are consistent because the employer is purchasing something the claimant of the current cohort does not sell: a reduction in the flow of *future* claims.

5.3 Summary of Results

Table 3: Summary of Equilibria across Model Extensions

Model	Key condition	Worker	Employer
Static, risk neutral	(4.9) and (4.10) hold	Dispute	Dispute
Static, risk neutral	(4.9) fails, (4.10) holds	Compose	Dispute
Static, risk neutral	(4.9) holds, (4.10) fails	Dispute	Compose
Static, risk neutral	neither holds	Compose	Compose
Risk aversion	$\text{CE} > (1-\phi)v_s$	Dispute	Dispute / Compose
Risk aversion	$\text{CE} \leq (1-\phi)v_s$	Compose	Dispute / Compose
Reputation	$\delta \geq \delta^*$ and (5.29)	File if $\varepsilon_i < \varepsilon^*(1)$	Tough, $\bar{q} = 1$
Reputation	$\delta \leq \delta_0$	File if $\varepsilon_i < \varepsilon^*(0)$	Soft, $\bar{q} = 0$
Reputation	$\delta_0 < \delta < \delta^*$, under (5.25)	File if $\varepsilon_i < \varepsilon^*(\bar{r})$	Interior $\bar{q} \in (0,1)$

In every row the filing rate is strictly positive (Lemma 5.3): deterrence is partial at every reputation the employer can attain. The static rows compare payoffs discounted at r_L and r_H over T; the risk-aversion rows compare undiscounted terminal wealth, so $\rho = 0$ in those rows recovers Section 4.1 and not Section 4.2. Adding the claimant's discount factor divides the left-hand side of the risk-aversion condition by $(1+r_L)^T$ and leaves Corollary 5.1 unchanged, since the sign of $\partial \text{CE}/\partial\rho$ is unaffected. The three blocks are alternative accounts of the same margin and are not combined here; see Section 6.2 and Section 6.4.

6 The Model and the Data

This section confronts the model of Sections 4 and 5 with the TRT-2 corpus at the level of institutional configuration and qualitative pattern. We do not attempt a quantitative calibration. Several of the model's objects require data work that is under way and not yet complete, and a calibration built on the present extraction would convey a precision the corpus does not support. What the corpus can do, and what we do with it below, is establish that the configuration the model assumes is the one that actually obtains, that the constraint forcing our definition of reputation is real, and that the model's state variable has a directly measurable counterpart. Section 6.2 then puts the two mechanisms of Section 5 to the corpus one at a time, asking of each whether these data can support it, and reporting where they cannot.

6.1 The institutional configuration the model assumes

Lemma 5.1 requires the claimant's out-of-pocket exposure to be zero in every state. Three Brazilian rules deliver that jointly, and each is a matter of law, verifiable independently of

our extraction.

The waiver of judicial fees (*gratuidade de justiça*) removes court costs and expert fees, setting $c_w = 0$. It is granted on a signed self-declaration of financial incapacity, which under Law 7,115/1983 carries a presumption of veracity, and it is decided by the same judge who decides the merits. Table 4 reports its incidence in our sample.

Table 4: Requests for Waiver of Judicial Fees

	Share of sample	Number of cases
Total sample	100.00%	129,720
Request for judicial fee waiver	85.02%	110,287
Waiver granted	84.70%	109,879
Granted, conditional on request	99.63%	

Source: authors' calculations.

Two figures matter and they are different figures. The approval rate conditional on requesting, 99.63%, shows that the waiver is granted essentially automatically once asked for. The unconditional share, 84.70%, is the fraction of claimants for whom $c_w = 0$, and it is the one that belongs in the model: roughly six claimants in seven, with the remaining one not represented by any type in our framework. Yeung and Bruscato (2023) document the same granting practice on a larger and more recent sample of first-instance cases.

Fee shifting does not expose the losing claimant to the employer's attorney fees. Under the case law governing our sample period, a losing labor claimant owed no such fees on mere *sucumbência*, whether or not she held a waiver. This is what allows the fee-shifting term to be set to zero in Remark 4.1, and it is a distinct rule from the waiver: the waiver removes costs owed to the court, not exposure to the opposing party.

Representation is contingent. The *quota litis* retainer is the ordinary form here: the attorney takes a share of what the claimant recovers and nothing when she recovers nothing. This is what makes ϕ a share rather than an outlay.

The conjunction is what matters. Any one of the three failing would leave the claimant with a state in which her terminal wealth falls below w_0, and then a sufficiently risk-averse claimant would not file. Together they produce a lottery whose worst realisation is a gain of zero, which is the precise content of Lemma 5.1 and the precise sense in which these institutions guarantee participation.

6.2 Two questions the corpus can be asked

Sections 5.1 and 5.2 are two alternative accounts of the same margin, not two halves of a single account. We therefore put each of them to the corpus separately, and record where the answer is negative.

6.2.1 Does claimant risk aversion, on its own, account for disputing?

It does not, and the model is what tells us so. Corollary 5.1 is exact and requires no approximation: a more risk-averse claimant has a lower certainty equivalent, so the region in which she prefers to compose is strictly larger. Risk aversion widens the set of mutually acceptable compositions rather than emptying it. As a standalone explanation of why

claims are adjudicated it therefore predicts the opposite of what it is invoked to explain, and no configuration of the data could rescue it: the sign is a theorem, not a calibration.

This is worth stating plainly because the opposite intuition is common. A vulnerable claimant facing a long and uncertain proceeding is easy to picture as someone who fights on. The model says she is the one who settles cheaply, and that the party with the reason to fight is the one that is risk neutral and patient.

What risk aversion does deliver is a candidate account of a different quantity, the *level* at which compositions are struck when they are struck. Proposition 5.1 prices a composition at the claimant's reservation value CE, so a composition that is low relative to the expected award implies a high ρ. That is a testable statement, and we cannot test it here. It requires composition amounts whose procedural stage is identified, which Section 3.2 explains this corpus does not supply, and an external measure of ρ or of r_L, which no part of a decision text contains. Any inversion of an observed composition into a value of ρ performed inside this model would be circular, because CE is what the model uses to price the composition in the first place.

Risk aversion also does work in Section 5.2, where a lower $\widehat{\text{CE}}$ raises $\eta(1)$ by (5.27) and makes a record of toughness more productive. That is a channel, not a result of this paper: we have not established that it is quantitatively important, and we do not claim it. Section 6.4 states what would settle it.

6.2.2 Does reputation account for the equilibrium the data show?

Here the answer is affirmative in a limited and precise sense: two features of the corpus are what the reputation model requires and are difficult for the alternatives to accommodate, while the model's central prediction is not observable in these data at all. We take the three in turn, and close with a qualitative pattern.

First, a reputation in this market cannot be made of victories, and the data are what establish that. Table 2 reports that employers are ordered to pay in 88.55% of adjudicated decisions. Read with the caveat on partial victories in Section 3.2, that figure delivers one parameter, p_c, and no more; in particular we do not read it as evidence against the selection hypothesis, for the reasons given in Section 2. What it does establish is the premise of Remark 5.4. Under a policy of disputing every claim an employer's win rate converges to $1 - p_c$, which these data put near 0.115, so an employer here cannot build a reputation out of victories however aggressively it litigates. This is not a modelling convenience. It is a constraint the data impose on what a reputation in this market can be made of, and it is why Definition 5.4 places the record over the employer's composition policy, which the employer does control.

Second, the phenomenon is a stationary regime and not a transition. The corpus covers sixteen years of continuous heavy filing against the same recurring defendants. A model in which deterrence operates through a threshold predicts an absorbing state in which filings against a tough defendant stop altogether, which is not what sixteen years of data show. Lemma 5.3 is the property that reproduces the observed pattern: because claimants differ in what waiting costs them, the filing rate is bounded away from zero at every reputation the employer can attain, and deterrence is partial by construction. Heterogeneous participation costs are not a technical device here; they are what allows persistent litigation to be an equilibrium rather than a phase.

Third, and against the two above, the central prediction is not observable in this corpus. Proposition 5.2(i) makes the state variable exactly one minus the defendant's composition rate. Compositions do not generate a decision, so cases resolved by agreement are absent

from a corpus built by mining decisions, by construction and not by oversight. We therefore cannot measure $\bar{r}$ for any defendant, and the premise that recurring defendants compose little is a hypothesis of this paper rather than a finding of it. The aggregate statistics point the other way for the branch as a whole: the Conselho Nacional de Justiça (2016) reports that the labor courts have the highest conciliation index in the Brazilian judiciary, with roughly two fifths of first-instance knowledge-phase sentences resolved by homologation of an agreement. That figure is an average over all defendants, most of whom appear once, and it is silent about the recurring defendants the model concerns. But it is the number a reader should have in front of them, and it makes the firm-level composition rate the first item on the agenda below rather than a refinement of it.

A qualitative pattern, reported as such. Among the defendants in our sample facing one hundred or more claims, both average awards and recorded composition amounts vary widely, and some defendants record no composition at all while others record amounts comparable to a substantial fraction of the average award. Dispersion in amounts is consistent with the sorting the model predicts, but it does not establish it: amounts scale with the size of the claim, the recorded figures are nominal and aggregated across 2001 to 2017 without deflation, and the procedural stage of the recorded agreements is not yet identified, so we cannot say whether they are the pre-judgment agreements the model prices. A firm-level test of the regime prediction requires composition *rates*, not amounts.

6.3 The model's objects and their observable counterparts

Table 5 maps the objects of the model that a corpus of decisions could in principle measure to the quantity that would measure them, and states what stands in the way at present. The claimant's risk aversion ρ, the employer's discount factor δ, the turnover rate λ and the distribution G are not in the table, because no part of a decision text bears on them.

Table 5: Model objects and their empirical counterparts

Object	Observable counterpart	Status
p_c	Share of adjudicated cases with a positive award	Measured at 88.55%; to be replaced by a value-weighted success rate
v_c	Average award, conditional on an award	Recorded; conditioning and deflation pending
v_s	Average composition amount	Recorded for a subset of defendants; the procedural stage of the recorded agreements is not yet identified
c_w, ϕ	Court costs and the contingent share	$c_w = 0$ established institutionally for 84.70% of claimants; ϕ is not observed in the corpus
T, t_s	Months to execution; months to the hearing	Available from published judicial statistics
$\bar{r} = \bar{q}$	One minus the defendant's composition rate	**Not observable in this corpus**; requires procedural movement records
$\eta(1)$, α	Responsiveness of filings to a defendant's record	Not identified by a corpus of decided cases (Remark 5.11)
κ	$1 - v_s^F/c$, from v_s, v_c, p_c, p_e, c_e, r_H, T and t_s	Computable once the items above are settled

Two entries deserve emphasis. The state variable of the model is not a latent construct: Proposition 5.2(i) makes $\bar{r}$ exactly one minus the defendant's composition rate, which is a number a court's case-management system records. That the state should be directly measurable is a virtue of the formulation, and that we cannot yet measure it is a limitation of the extraction rather than of the theory. The deterrence elasticity, by contrast, is not measurable from decided cases at all, for the reason given in Remark 5.11, and any quantitative statement about δ^* will rest on an assumption about G that should be reported as such.

6.4 What would discipline the model

We close with the ordered list of what a quantitative version of this section requires.

1. **Procedural movement records**, which deliver composition rates and hence $\bar{r}$ directly, and which also settle at what stage the recorded agreements were struck.
2. **Validation of the extraction**, on the protocol described in Section 3.2.
3. **A value-weighted success rate**, replacing the binary indicator and with it the interpretation of p_c.
4. **Deduplication to the case level and deflation** to a common base year.

5. **An external measure of the claimant's discount rate or of** ρ, either of which pins down CE independently of the model and makes the pricing statement of Proposition 5.1 testable without circularity.

6. **The 2017 labor reform and its partial reversal in 2021**, which moved the claimant's out-of-pocket exposure in opposite directions at two known dates and offer the identification this literature currently lacks.

Two of these deserve a word about what they are for. This paper has kept claimant risk aversion and employer reputation apart on purpose, because each is a complete account of the disputing margin on its own and the two answer to different evidence: risk aversion is disciplined by composition *amounts*, reputation by composition *rates*. Items 1 and 5 are exactly what a joint model would need. With firm-level composition rates and an external measure of ρ, the two channels can be identified separately and then allowed to interact, since a lower $\widehat{\text{CE}}$ raises $\eta(1)$ by (5.27) and therefore lowers δ^*: a defendant facing more risk-averse claimants would need less patience to find permanent toughness worthwhile. That is a prediction about the interaction of the two mechanisms and we regard it as the natural sequel to this paper, not as one of its results. We flag it here rather than developing it because the present data cannot separate the two channels, and a joint model estimated on data that identify neither would be an exercise in fitting rather than in testing.

7 Conclusions

This paper set out to explain why an employer that is ordered to pay in the large majority of the claims it defends nevertheless declines to compose, claim after claim, when composition is available and cheap.

Our answer is that the employer is not buying a cheaper resolution of the present claim. It is buying a public record of refusing to compose, which lengthens the expected duration of any claim brought against it and thereby thins the flow of claims it will face in future. The mechanism requires neither divergent beliefs about the merits nor private information about them. It requires only that adjudication be slow relative to conciliation, and that claimants differ in what waiting costs them.

Three results carry the argument. A reputation for toughness deters entry if and only if the ratio of the two values at stake is smaller than the ratio of the two delays, each measured as a discounted annuity (Lemma 5.2); the condition is easy to satisfy when adjudication is slow, and automatic when compositions price at the claimant's reservation value. Because claimants differ in their participation cost, deterrence is always partial and the filing rate is bounded away from zero at every reputation the employer can attain (Lemma 5.3); the stationary outcome is persistent litigation rather than its disappearance, which is the phenomenon we set out to explain and which a model with homogeneous claimants cannot reproduce. And permanent toughness requires the employer's patience to exceed a threshold that falls with the responsiveness of entry and with the speed at which the market's record forms (Proposition 5.2), and beats composing every claim when the deterrence dividend exceeds $1/(1-\kappa)$ (Lemma 5.5 and Corollary 5.4).

We are deliberate about how the three forces of the paper relate, because a more convenient account of it would be wrong. The discount-rate wedge compresses the set of mutually acceptable compositions; claimant risk aversion, taken alone, *widens* it, and

therefore cannot be the reason claims are adjudicated; and the reputational mechanism makes the employer refuse compositions that the one-shot comparison would have it accept. They do not push in the same direction on the same margin, and we have not combined them. Section 6.2 puts each to the corpus separately and reports where the answer is negative, and Section 6.4 states what a joint model would require before it could be estimated rather than merely written down.

We are equally deliberate about what the Brazilian evidence supports. The corpus establishes the institutional configuration the model assumes: a court-fee waiver granted essentially automatically and held by roughly six claimants in seven, a fee-shifting rule that leaves the losing claimant unexposed, and contingent representation. Together these produce a lottery with no losing ticket, which is what guarantees participation at any degree of risk aversion. The corpus also establishes the constraint that forces our definition of reputation, since an employer that disputes everything still loses most of what it disputes and so cannot build a record of victories. What the corpus does not yet establish is any magnitude, nor the premise itself. Composition rates are not observable in a corpus of decisions, the extraction is not yet validated, and the deterrence elasticity is not identified by decided cases at all; and the aggregate conciliation statistics for the labor branch, reported in Section 6.2.2, are a caution rather than a confirmation. A quantitative calibration therefore awaits the data work set out in Section 6.4, and we prefer to state the model's empirical content as a list of measurable objects rather than to report numbers the present extraction cannot support.

Three extensions seem to us most promising. The 2017 labor reform charged losing claimants the employer's fees and the Supreme Court struck that provision down in 2021, moving the claimant's out-of-pocket exposure in opposite directions at two known dates; this is the identification the literature lacks, and our framework says precisely which object it identifies. An external measure of either the claimant's discount rate or her risk aversion would pin down CE independently of the model and turn a qualitative statement into a test, and together with firm-level composition rates it would allow the two mechanisms of this paper to be identified separately and then combined. And the mechanism should operate wherever adjudication is slow, the defendant is a repeat player and the plaintiff is not, which points to consumer and tax litigation as the natural next settings.

References

Acciarri, H. A., & Garoupa, N. (2013). On the judicial interest rate: Towards a law and economic theory. *Journal of European Tort Law*, *4*(1), 34–62. https://doi.org/10.1515/jetl-2013-0002

Bebchuk, L. A. (1984). Litigation and settlement under imperfect information. *The RAND Journal of Economics*, *15*(3), 404–415.

Che, Y.-K., & Yi, J. G. (1993). The role of precedents in repeated litigation. *Journal of Law, Economics, and Organization*, *9*(2), 399–424.

Conselho Nacional de Justiça. (2016). Justiça em números 2016: Ano-base 2015 [First edition to report the conciliation index].

Conselho Nacional de Justiça. (2018). *Justiça em números 2018: Ano-base 2017* (tech. rep.). CNJ. Brasília.

Conselho Nacional de Justiça. (2025). *Justiça em números 2025* (tech. rep.). CNJ. Brasília.

Cooter, R. D., & Rubinfeld, D. L. (1989). Economic analysis of legal disputes and their resolution. *Journal of Economic Literature*, *27*(3), 1067–1097.

Daughety, A. F., & Reinganum, J. F. (1999). Hush money. *The RAND Journal of Economics*, *30*(4), 661–678.

Daughety, A. F., & Reinganum, J. F. (2005). Economic theories of settlement bargaining. *Annual Review of Law and Social Science*, *1*, 35–59.

Eisenberg, T. (1990). Testing the selection effect: A new theoretical framework with empirical tests. *The Journal of Legal Studies*, *19*(2), 337–358.

Fudenberg, D., & Levine, D. K. (1989). Reputation and equilibrium selection in games with a patient player. *Econometrica*, *57*(4), 759–778.

Galanter, M. (1974). Why the "haves" come out ahead: Speculations on the limits of legal change. *Law and Society Review*, *9*(1), 95–160.

Gould, J. P. (1973). The economics of legal conflicts. *The Journal of Legal Studies*, *2*(2), 279–300.

Hubbard, W. H. J. (2018, February). *Stalling, conflict, and settlement* (Working Paper) (SSRN Working Paper 3127761). University of Chicago Law School, Coase-Sandor Institute for Law and Economics. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3127761

Klerman, D., & Lee, Y.-H. A. (2014). Inferences from litigated cases. *The Journal of Legal Studies*, *43*(2), 209–248.

Kreps, D. M., & Wilson, R. (1982). Reputation and imperfect information. *Journal of Economic Theory*, *27*(2), 253–279.

Landes, W. M. (1971). An economic analysis of the courts. *The Journal of Law and Economics*, *14*(1), 61–107.

Mailath, G. J., & Samuelson, L. (2001). Who wants a good reputation? *The Review of Economic Studies*, *68*(2), 415–441.

Mailath, G. J., & Samuelson, L. (2006). *Repeated games and reputations: Long-run relationships*. Oxford University Press.

Milgrom, P., & Roberts, J. (1982). Predation, reputation, and entry deterrence. *Journal of Economic Theory*, *27*(2), 280–312.

Miller, G. P. (1994). Economic analysis of civil procedure: Introduction. *The Journal of Legal Studies*, *23*(1), 303–306.

Pinheiro, A. C. (2000). *Judiciário e economia no brasil*. Editora Sumaré.

P'ng, I. P. L. (1983). Strategic behavior in suit, settlement, and trial. *The Bell Journal of Economics*, *14*(2), 539–550.

Posner, R. A. (1973). An economic approach to legal procedure and judicial administration. *The Journal of Legal Studies*, *2*(2), 399–458.

Priest, G. L., & Klein, B. (1984). The selection of disputes for litigation. *The Journal of Legal Studies*, *13*(1), 1–55.

Ramseyer, J. M., & Rasmusen, E. B. (2013). Are americans more litigious? In F. H. Buckley (Ed.), *The american illness: Essays on the rule of law*. Yale University Press.

Reinganum, J. F., & Wilde, L. L. (1986). Settlement, litigation, and the allocation of litigation costs. *The RAND Journal of Economics*, *17*(4), 557–566.

Shavell, S. (1982). Suit, settlement, and trial: A theoretical analysis under alternative methods for the allocation of legal costs. *The Journal of Legal Studies*, *11*(1), 55–81.

Shavell, S. (1996). Any frequency of plaintiff victory at trial is possible. *The Journal of Legal Studies*, *25*(2), 493–501.

Spier, K. E. (1992). The dynamics of pretrial negotiation. *The Review of Economic Studies*, *59*(1), 93–108.

Spier, K. E. (2007). Litigation. In A. M. Polinsky & S. Shavell (Eds.), *Handbook of law and economics* (pp. 259–342, Vol. 1). Elsevier.

Wollschläger, C. (1998). Exploring global landscapes of litigation rates. In J. Brand & D. Stempel (Eds.), *Soziologie des rechts: Festschrift für erhard blankenburg zum 60. geburtstag*. Nomos.

Yeung, L. L.-T., & Bruscato, A. (2023). *Gratuidade e acesso à justiça* (tech. rep.). Conselho Nacional de Justiça and Insper. Brasília. https://www.cnj.jus.br/wp-content/uploads/2023/11/pesq-gratuidade-insper.pdf